HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

Cardio Digital Twin

A twin that watches heart patients between clinic visits, and tells the doctor who needs attention today

App: DilSe ("dil se", Hindi for "from the heart")

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Team One Fold · solo entry · Parul University

THE PROBLEM

The gap between clinic visits

Seen only now and then

One ECG at the clinic. Then silence until the next visit.

Daily data nobody reads

Smartwatches record heart rate, sleep and steps. No one can watch thousands of patients.

One rule fits no one

"Heart rate above 100" ignores that every person has their own normal.

The challenge: combine clinic data with daily wearable data, and flag who is at risk.

HEALTHCARE USE CASE

Remote follow-up in four steps

1

Clinic ECG sets the risk

The AI reads the ECG: high risk or low risk.

2

Daily readings vs own normal

Heart rate, HRV, sleep and steps.

3

Risk tunes the alarm

High-risk patients get an earlier alarm.

4

Morning worklist

Critical, Concern or Stable, with a reason and a next step.

It supports triage only: no diagnosis, no treatment advice. The doctor decides.

THE DATA

Two streams: one real, one simulated

STREAM 1 · CLINIC ECG · REAL

21,799 ECGs from 18,869 patients

- PTB-XL, labelled by cardiologists
- Open and anonymised
- Split by patient: no one in both train and test

STREAM 2 · WEARABLE · SIMULATED

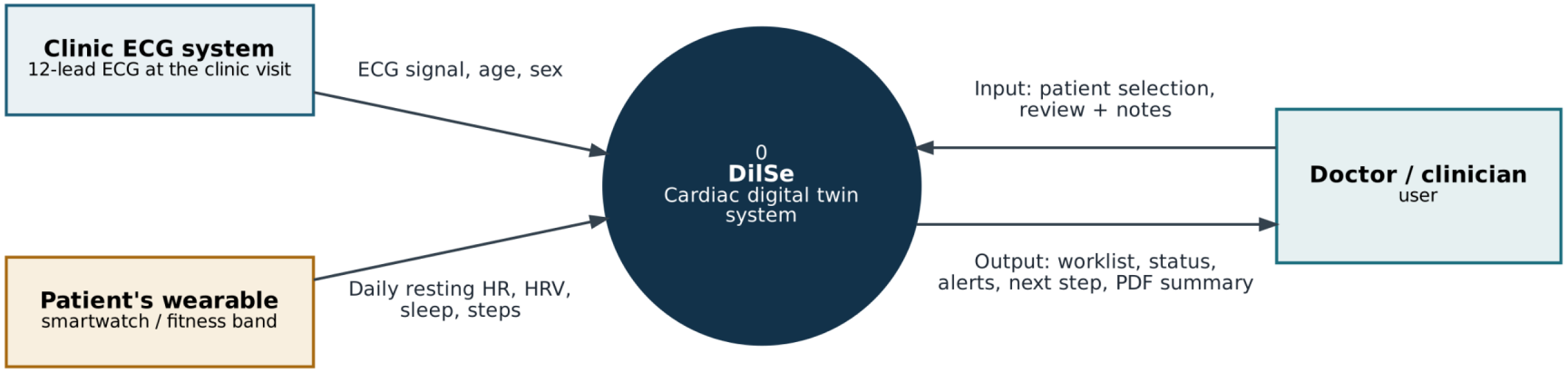
60 days × 4 measures per patient

- Settings from real data (PMData, MMASH)
- Sees only age and sex, never the ECG result
- Half the patients get a deterioration episode

Real where it matters most. Simulated only where no open data exists.

DATA FLOW DIAGRAM · LEVEL 0

What goes in, what the doctor gets

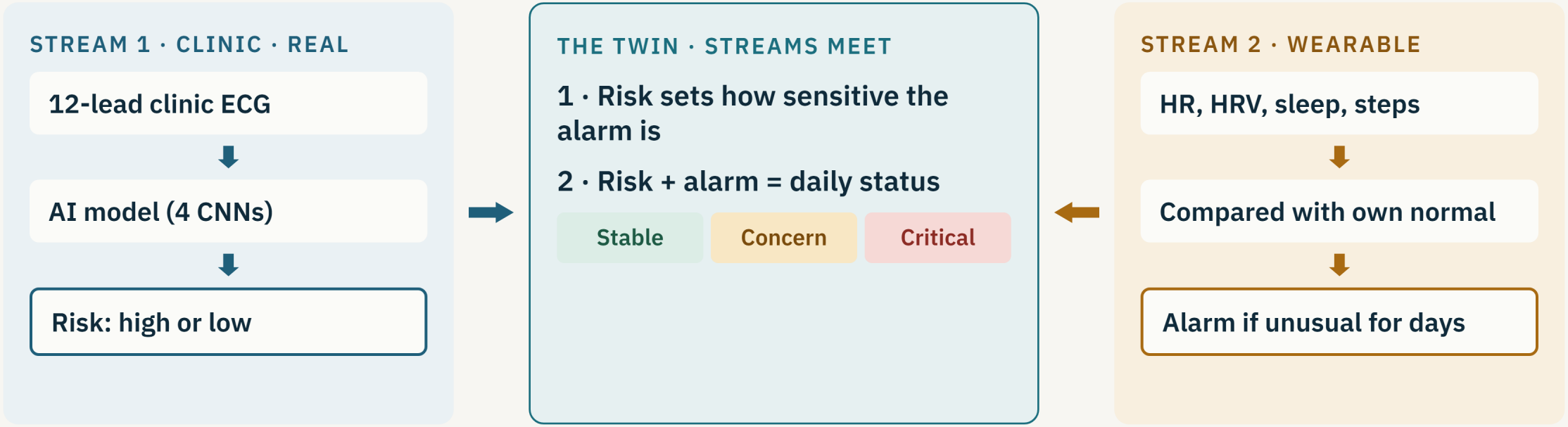


2 sources in | **1 process: DilSe** | **1 user: the doctor**

Two data streams in. One clear answer out: who needs attention, why, and what to do.

SYSTEM ARCHITECTURE

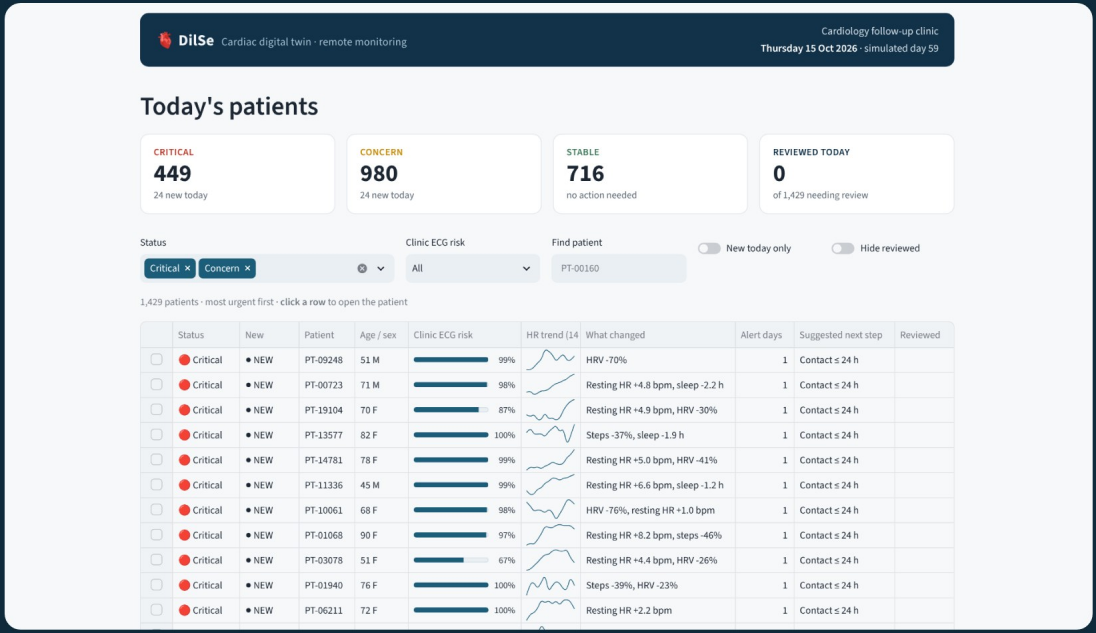
Two streams, one daily status

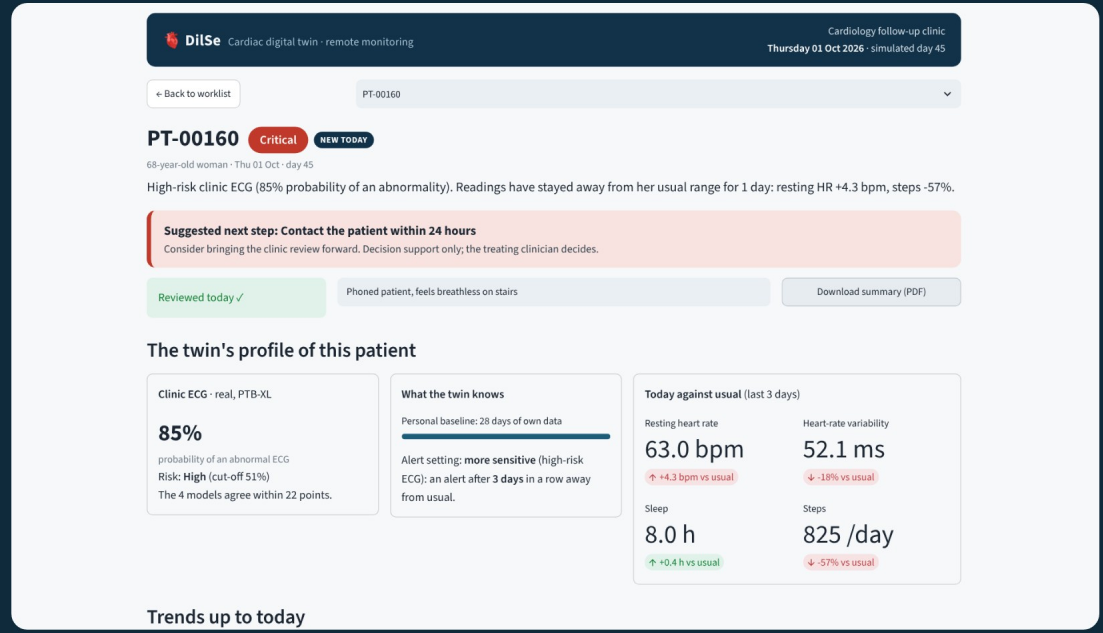


DilSe app: every morning, a worklist with a reason and a next step for each patient.

DILSE APP · FOR THE DOCTOR

Who needs attention today, and why





Left: the morning worklist. Right: one patient, with a summary, a next step and a PDF.

STREAM 1 · ECG MODEL

An AI that reads the raw ECG

12-lead ECG
10 seconds, raw signal

→

4 small neural networks
1-D CNN, 190,000 parameters each

→

Is it abnormal?
A probability, 0 to 100%

DESIGN DECIDED BY EXPERIMENT, RULE WRITTEN FIRST

100 Hz worked as well as 500 Hz, 5× faster

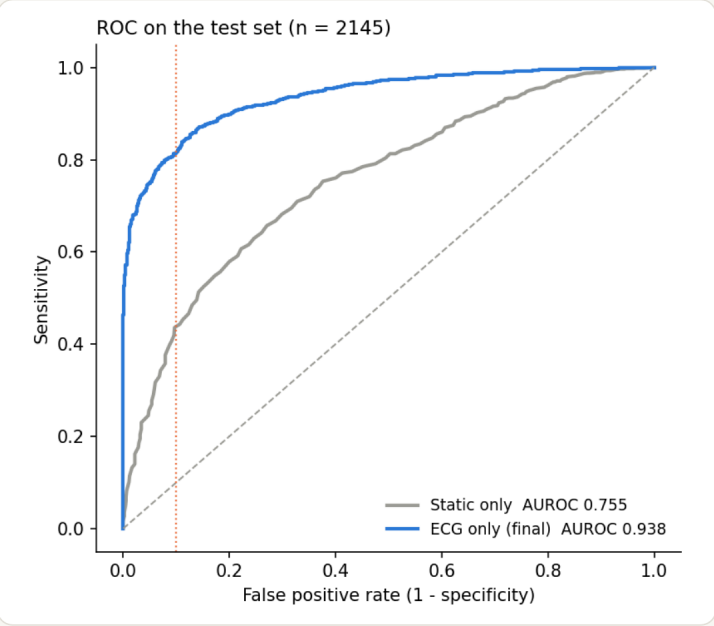
Bigger ResNet did not clear the pre-set bar

Age and sex added nothing: ECG only

Small enough to train on a laptop. High risk if the score is 51% or more.

RESULTS · ECG MODEL · REAL DATA

Tested once, on 2,145 unseen patients



0.938

AUROC with the ECG. Age and sex alone: **0.755**.

80%

of abnormal ECGs caught

91%

of normal ECGs correctly cleared

Pick one abnormal and one normal ECG: 94% of the time, the model ranks the abnormal one higher.

Cardio Digital Twin · 9 · 95% CI 0.928–0.947 · validation AUROC 0.942 · test set scored once by a self-locking script

STREAM 2 · WEARABLE SIMULATION

Simulated, but every number has a source

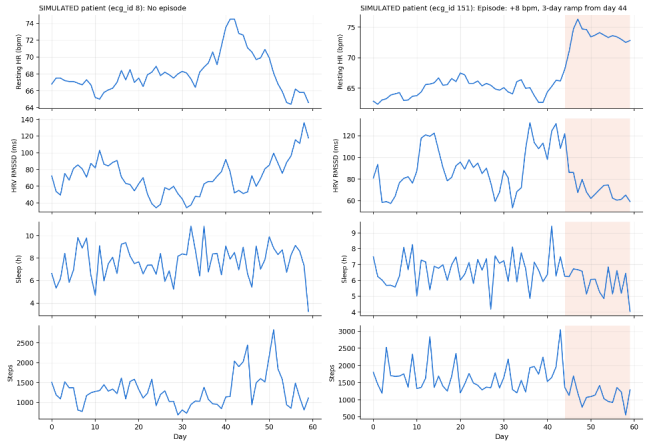
Each day = the person's usual level + a weekday effect + a day-to-day wobble with memory

Wobble: fitted on real Fitbit users (PMData)

Heart rate: Quer et al. 2020

Steps: India average, 4,297 a day (Althoff 2017)

Episodes: in half the patients, heart rate +3 to +12 bpm



No leakage: the simulator sees only age and sex, never the ECG result.

PERSONAL-BASELINE MONITOR

Compared with their own normal

Every day, the twin asks:

1 Is today unusual for this patient?

2 Is it in the unhealthy direction?

3 Has it lasted several days?

Their normal = their own last 4 weeks, skipping the most recent week

Episodes caught (validation)

Simple rule: heart rate +10 bpm for 3 days



28.8%

DilSe: own-normal monitor



90.0%

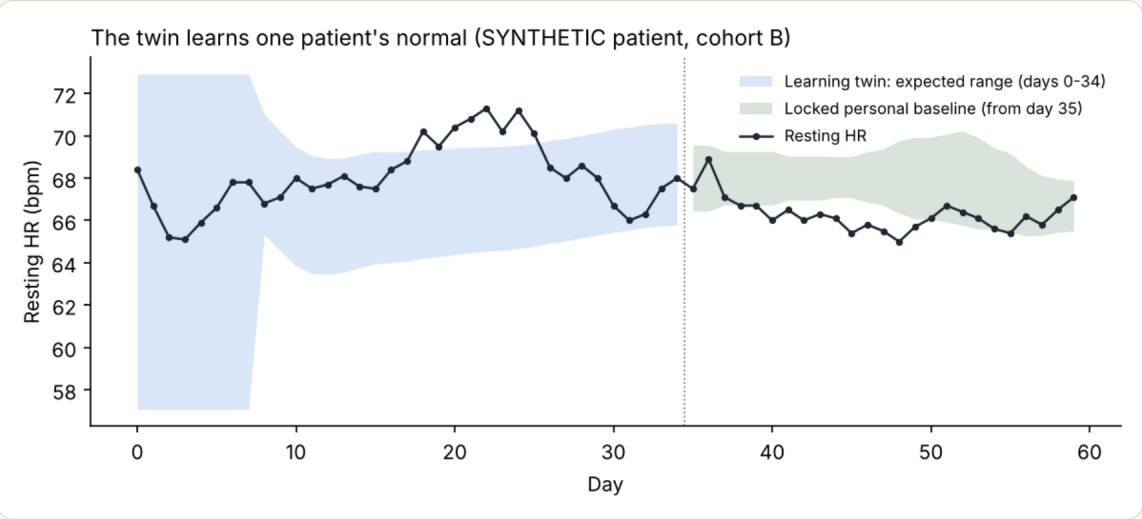
Same false-alarm budget: 0.5 per patient per month

Comparing each patient with their own normal catches 3× more than a simple rule.

Cardio Digital Twin · 11 · +5 bpm episodes: 93% vs 85% with a fixed baseline · real people (PMData): 0.63 alarms per month

THE TWIN LEARNS FROM DAY ONE

Monitoring from day 0, not day 35



Day 0: ±8 bpm from population values. Day 10: under ±3 bpm, from the patient's own data.

11% → 87%

early episodes caught within two weeks

False alarms within budget: 0.49 per patient-month

Real people (PMData): alarm rate within limits

A new patient is watched from the first day, not from week five.

THE TWIN · COMBINING BOTH STREAMS

ECG risk sets how closely we watch

	No alarm	Alarm
Low risk	Stable	Concern
High risk	Concern	Critical

Risk tunes the alarm

High risk: alarm after 3 days
Low risk: alarm after 4 days

On test patients

High-risk patients caught: **93.9%** (vs 92.4%), at the same false-alarm rate

The ECG decides how closely we watch. The wearable decides when to act.

INTEGRITY

How the results were kept honest

Rules before runs

28 logged decisions, each rule written before the experiment ran.

Test set opened once

All tuning on validation. The final script locks itself after one run.

13 automated checks

No patient in two splits, no look-ahead, settings unchanged.

Failures reported too

A heart-age model failed its rule: left out. One early test peek: disclosed.

Rules first, test once, report what failed.

LIMITATIONS

What this prototype does not show yet

Limitation	In plain words
Now, not the future	The ECG model finds a current abnormality; it does not forecast events
German data	PTB-XL is from Germany; performance on Indian patients is unknown
Some misses	About 1 in 5 abnormal ECGs is missed (242 of 1,236)
Simulated wearables	All wearable data is simulated; real people had 1.3× the alarm budget
Learning twin	Its synthetic early-detection gain is optimistic
Static data	Only ECG, age and sex; no lab results or diagnosis history

A research prototype, not a medical device. No treatment advice.

NEXT STEPS

From prototype to clinical pilot

1 Validate the ECG model on Indian patients' ECGs with a partner hospital

2 Replace the simulated stream with real paired ECG + wearable data

3 Link to outcomes, to predict future events

4 Add lab results and diagnosis history

Thank you · Priyanka Bharwani